



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

ARTIFICIAL INTELLIGENCE [ELE-4409]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) With a neat schematic explain the typical model used in artificial intelligence system? (5)
A)
B) Discuss the concept of rationality. Discuss the parameters on which rationality depends? (5)
- 2) Discuss the concept of breadth first search algorithm and develop its pseudocode? (5)
A)
B) Discuss on Heuristic functions used in solving problems? (5)
- 3) Discuss adversarial search problem? Explain how it is different from other search problems? (5)
A)
B) Discuss game tree? Explain a typical game tree for tic-tac-toe problem? (5)
- 4) Explain the concept of objects, relation, and functions with respect to natural language? Demonstrate them with an example? (5)
A)
B) Discuss in brief the steps used in knowledge engineering process? (5)
- 5) Discuss the concept of exhaustive decomposition and partition with an example? (5)
A)
B) Discuss the concept of upper ontology. Explain the characteristics of general-purpose ontologies? (5)

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